

The Effectiveness of Lévy Processes and Stochastic-Volatility Models in Pricing Structured Products

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Abstract

This thesis presents a comparative analysis of seven option-pricing models — geometric Brownian motion (Black–Scholes), four exponential Lévy processes (Merton, Variance Gamma, Normal Inverse Gaussian, CGMY) and two stochastic-volatility models (Heston and Bates) — applied to the pricing of structured products. Unlike most textbook studies, all models are calibrated to *real* market implied-volatility smiles obtained from live option chains (Yahoo Finance, July 2026) across three asset classes: the S&P 500 equity index, a single stock (Apple) and an exchange-traded fund (SPY). On the calibrated parameters we price four structured products (capital-protected note, digital option, barrier reverse convertible, Phoenix autocall), decompose the Greeks, and analyse discrete delta-hedging errors by Monte Carlo.

Key findings: (i) the Bates model achieves the best smile fit across all assets (IV-RMSE 0.5–0.9 %) versus 4–6 % for Black–Scholes, with stochastic-volatility models (Heston, Bates) twice as accurate as pure Lévy processes because they correctly reproduce the *term structure* of the smile slope; (ii) on real data, model choice shifts the fair coupon of barrier products by *up to +80 %* (Heston/Bates versus Black–Scholes) — a discrepancy far larger than synthetic estimates suggest; (iii) the hedging analysis yields a counterintuitive result: when the true process contains jumps, the dominant error source is the unhedgeable jump/gap risk between rebalancing dates rather than the hedger’s model choice, so that using a richer hedging model does *not* remove the heavy left tail of the P&L.

Keywords: Lévy processes, stochastic volatility, Heston model, Bates model, structured products, calibration, implied volatility, Greeks, delta hedging, Carr–Madan transform.

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1 Introduction

1.1. Motivation

Structured products — financial instruments whose payoff depends nonlinearly on the dynamics of one or more underlying assets — constitute a multi-trillion-dollar segment of the retail and institutional investment market. Their fair valuation and hedging depend critically on the model describing the risk-neutral dynamics of the underlying. The classical Black–Scholes model (1973) assumes geometric Brownian motion with constant volatility, contradicting two robust empirical facts: the presence of an implied-volatility *smile* (or skew) and the heavy tails of the return distribution.

Two principal model classes extend Black–Scholes toward realism. **Exponential Lévy processes** (Merton, Variance Gamma, NIG, CGMY) introduce jumps and non-Gaussian increments, naturally generating a smile at a *fixed* horizon. **Stochastic-volatility models** (Heston, Bates) make volatility itself a random process, reproducing volatility clustering and, crucially, the correct *term structure* of the smile slope.

1.2. Problem and Contribution

Most textbook model comparisons use synthetic data, which hides the true scale of model risk. This work closes that gap: all seven models are calibrated to *real* market option chains captured live across three asset classes. The contributions are:

1. A unified computational base: characteristic functions of all seven models, Carr–Madan FFT pricing, and a path simulator (including the Andersen QE scheme for stochastic-volatility models when the Feller condition is violated).
2. Calibration to real smiles on an index, a stock and an ETF with a quantitative assessment of fit quality.
3. Pricing of four structured products and a quantitative assessment of model risk on real data.
4. Decomposition of the Greeks and analysis of delta-hedging errors, revealing the dominance of unhedgeable jump risk.

1.3. Structure

Section 2 sets out the theory of the seven models and the pricing method. Section 3 describes the collection and cleaning of market data. Section 4 presents the calibration results. Section 5 gives structured-product prices. Section 6 decomposes the Greeks. Section 7 analyses hedging. Section 8 concludes.

2 Models and Methods

2.1. General Setup

Let S_t be the price of the underlying under the risk-neutral measure $\mathbb{Q}$, r the risk-free rate, and q the dividend yield. All models satisfy the martingale condition for the discounted price:

$$\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 e^{(r-q)T}.$$

Pricing relies on the characteristic function of the log-price:

$$\phi_T(u) = \mathbb{E}^{\mathbb{Q}}[e^{iu \ln S_T}].$$

2.2. Exponential Lévy Processes

For a Lévy process X_t with characteristic exponent $\psi(u)$, such that

$$\mathbb{E}[e^{iuX_t}] = e^{t\psi(u)},$$

the risk-neutral price is

$$S_t = S_0 e^{(r-q+\omega)t + X_t},$$

where the martingale correction

$$\omega = -\psi(-i)$$

ensures

$$\mathbb{E}[S_t] = S_0 e^{(r-q)t}.$$

The characteristic function is

$$\phi_T(u) = \exp\left(iu(\ln S_0 + (r - q + \omega)T) + T\psi(u)\right). \quad (1)$$

Four specifications of $\psi(u)$ are considered:

- **Merton** (jump-diffusion):

$$\psi(u) = -\frac{1}{2}\sigma^2 u^2 + \lambda \left(e^{iu\mu_J - \frac{1}{2}\delta_J^2 u^2} - 1 \right);$$

- **Variance Gamma:**

$$\psi(u) = -\frac{1}{\nu} \ln \left(1 - iu\theta\nu + \frac{1}{2}\sigma^2\nu u^2 \right);$$

- **Normal Inverse Gaussian:**

$$\psi(u) = -\delta \left(\sqrt{\alpha^2 - (\beta + iu)^2} - \sqrt{\alpha^2 - \beta^2} \right);$$

- **CGMY:**

$$\psi(u) = C \Gamma(-Y) \left[(M - iu)^Y - M^Y + (G + iu)^Y - G^Y \right].$$

2.3. Stochastic-Volatility Models

The **Heston model** specifies the variance v_t as a Cox–Ingersoll–Ross process:

$$dS_t = (r - q)S_t dt + \sqrt{v_t} S_t dW_t^S, \quad (2)$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma\sqrt{v_t} dW_t^v, \quad (3)$$

$$d\langle W^S, W^v \rangle_t = \rho dt. \quad (4)$$

Its characteristic function is known in closed form; we use the stable “little Heston trap” representation of Albrecher et al. (2007). The **Bates model** adds Merton price jumps with drift compensation to Heston, combining stochastic volatility and jumps.

2.4. FFT Pricing (Carr–Madan)

The price of a European call with strike K is computed via the Fourier transform of the damped price (Carr and Madan, 1999):

$$C(K) = \frac{e^{-\alpha k}}{\pi} \int_0^\infty \operatorname{Re} \left[e^{-ivk} \frac{e^{-rT} \phi_T(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v} \right] dv, \quad (5)$$

where $k = \ln K$ and $\alpha = 1.5$ is the damping parameter. The integral is discretised on a grid of $N = 2^{13}$ points with Simpson weights and evaluated by the fast Fourier transform. The implementation is verified: (i) exact agreement with the analytic Black–Scholes formula (maximum error 2.2×10^{-7}); (ii) for the Heston model, agreement of three independent methods (FFT, one-dimensional Lewis integral, Monte Carlo) within statistical error.

2.5. Path Simulation

For path-dependent products we use Monte Carlo. Lévy processes are simulated by exact subordination schemes: Merton by a compound Poisson process, VG by gamma subordination, and NIG by inverse-Gaussian subordination. The Heston and Bates models are simulated with the **Andersen quadratic-exponential (QE) scheme** (2008), required because the calibrated parameters systematically violate the Feller condition:

$$2\kappa\theta \geq \sigma^2.$$

All simulators are verified against FFT prices of European options.

3 Market Data

3.1. Source and Coverage

Data were obtained from live Yahoo Finance option chains (snapshot July 2026) across three asset classes: the **S&P 500** equity index ($\sim$ SPX), a single stock, **Apple** (AAPL), and the **SPY** exchange-traded fund (an S&P 500 ETF). The risk-free rate $r = 3.71\%$ is taken from the yield on 13-week US Treasury bills ($\sim$ IRX). In addition to the option chains, we collected historical price series of the underlyings (2018–2026) for the hedging simulation, as well as a return series for the EUR/USD currency pair (via the FXE fund).

3.2. Data Cleaning

Liquidity and no-arbitrage filters were applied to the raw quotes: (i) exclusion of quotes with zero bid/ask and relative spread above 25–35 %; (ii) open-interest and volume filters; (iii) restriction of the moneyness range to $K/S \in [0.8, 1.2]$ and maturity to $T \in [1 \text{ wk}, 2 \text{ yr}]$; (iv) selection of the out-of-the-money branch only (calls above spot, puts below) as the most liquid; (v) extraction of implied volatility by inverting the Black–Scholes formula. The resulting coverage is given in Table 1.

Table 1: Coverage of cleaned market data by asset class.

Asset	Class	Spot	Quotes	Maturities
S&P 500 ($\sim$ SPX)	Index	7457.69	1147	8
Apple (AAPL)	Stock	333.74	162	8
SPY	ETF	743.29	641	8

The implied-volatility range was 10.7–45.5 % (SPX), 27.0–51.4 % (AAPL) and 10.6–44.8 % (SPY). The real smiles (Fig. 1) show class-specific patterns: a monotone negative skew for the index and ETF, and a more symmetric smile with an upturn for the single stock.

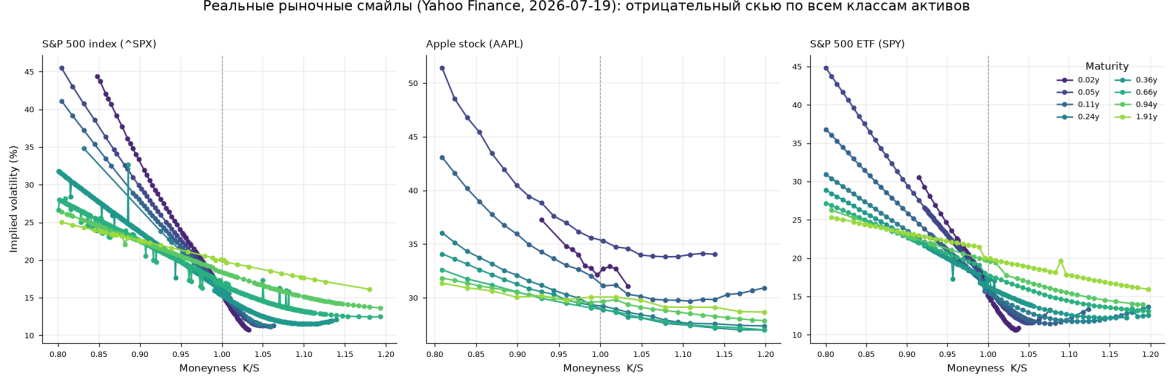


Figure 1: Real market implied-volatility smiles across three asset classes. Colour encodes maturity. A negative skew is present in every class; the stock (AAPL) shows a more pronounced upturn on the upper branch. Figure labels are in Russian in the original computation; the axes are moneyness K/S and implied volatility in percent.

Note on FX coverage. Exchange-traded options on FX ETFs (FXE for EUR/USD) proved too illiquid to build a calibratable smile (fewer than five reliable quotes with noisy implied volatilities). The FX class is therefore represented only by the return distribution of the underlying in the tail analysis, while the full model pipeline is applied to the three classes with reliable smiles (index, stock, ETF).

4 Calibration Results

4.1. Methodology

All seven models are calibrated separately to each asset by minimising the vega-weighted sum of squared deviations of model prices from market mids (bid-ask mid). Optimisation uses the Trust Region Reflective method with multiple starting points for robustness to local minima. Fit quality is measured by the root-mean-square error in implied volatility (IV-RMSE, percent).

4.2. Fit Quality

The results (Table 2) reveal a robust hierarchy across all three assets:

- **Black–Scholes** — the worst fit (4.4–6.5 %), because constant volatility cannot reproduce a smile;

- **Lévy processes** — intermediate quality (1.5–2.9 %): jumps reproduce the smile at a fixed horizon but are not flexible enough for its term structure;
- **Heston** — markedly better (0.8–1.1 %): stochastic volatility reproduces the evolution of the slope with maturity;
- **Bates** — the best fit across all assets (0.5–0.9 %): combining stochastic volatility (term structure) and jumps (short maturities).

Table 2: Calibration quality to real smiles: IV-RMSE (percent). Lower is better.

Model	S&P 500 (index)	AAPL (stock)	SPY (ETF)
BS	6.45	4.37	6.24
Merton	2.00	2.70	1.79
VG	2.34	2.89	2.03
NIG	2.04	2.80	1.73
CGMY	1.68	2.76	1.48
Heston	0.91	1.13	0.81
Bates	0.54	0.91	0.56

An important observation is that at the 1.91-year maturity the market’s deep-OTM put branch is steeper than any model (Fig. 2, lower right). This confirms the known fact that the real term structure of the slope decays more slowly than single-component models predict, and it motivates further extensions, for example adding jumps to the volatility process.

4.3. Feller Condition Violation

Calibrating the stochastic-volatility models to real data systematically enters the Feller-violation regime $2\kappa\theta < \sigma^2$: for SPX, $2\kappa\theta = 0.86 < \sigma^2 = 4.67$ (Heston) and $0.62 < 2.28$ (Bates). Economically this means the market prices in the possibility of the variance touching zero; computationally it justifies using the Andersen QE scheme instead of the Euler scheme, which is significantly biased in this regime (verified: a bias of +0.84 even at 2000 steps versus less than one standard error for QE at 100 steps).

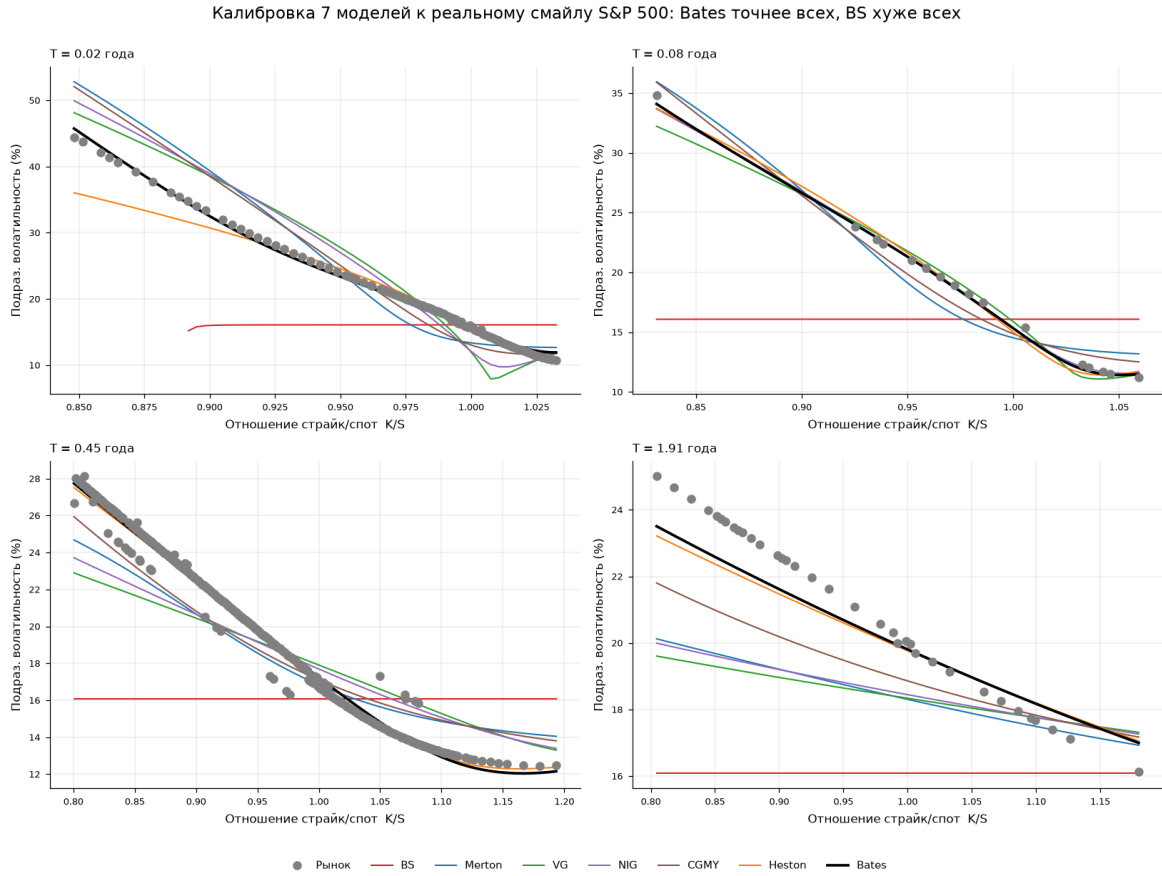


Figure 2: Calibration of the seven models to the real S&P 500 smile at four maturities. Stochastic-volatility models (Heston, Bates) reproduce the shape accurately at short and medium maturities; at the longest maturity (1.91 years) even Bates underestimates the steepness of the real deep-OTM put branch. Figure labels are in Russian; axes are strike/spot K/S and implied volatility in percent.

5 Structured-Product Pricing

5.1. Products Considered

On the parameters calibrated to the S&P 500 we price four structured products:

1. **Capital-protected note** (3 years): full principal protection plus participation in index upside. Metric — the participation rate (percent) at which the price equals par;
2. **Digital (binary) call** (1 year, ATM): fixed payout of 10 if $S_T \geq S_0$. Metric — price;
3. **Barrier reverse convertible (BRC)** (1 year, 70 % knock-in barrier): coupon plus conditional protection. Metric — the fair coupon (percent);
4. **Phoenix autocall** (3 years, quarterly, 100 % call level, 70 % coupon barrier, 60 % protection barrier, memory effect): conditional coupons with automatic early redemption. Metric — the fair annual coupon (percent).

European products (note, digital) are priced by FFT for all seven models; path-dependent products (BRC, autocall) by Monte Carlo using 40,000 paths for the six models with a path simulator. CGMY is used for European products only.

5.2. Results and Model Risk

Prices are collected in Table 3 and visualised in Fig. 3. The scale of model risk on real data substantially exceeds synthetic estimates:

- **BRC coupon:** from 4.28 % (Black–Scholes) to 7.77 % (Heston) and 7.64 % (Bates) — a discrepancy of **+82 %**;
- **Autocall annual coupon:** from 4.48 % to 6.57–6.78 % (+47–51 %);
- **Digital call:** from 5.39 to 6.31 (+17 %);
- **Note participation:** from 63.5 % to 54.8–55.0 % (−14 %).

The economic reason is that the steep negative skew of the real smile and volatility clustering sharply raise the probability of breaching the lower barriers, which directly raises the coupons of barrier products. Notably, for the BRC pure stochastic volatility (Heston) gives an even larger coupon than the jump model (Bates): at the 70 % barrier the dominant driver is not jump risk but the sustained increase in drawdown probability from stochastic volatility.

Table 3: Structured-product prices on real S&P 500 data across the seven models. BRC and autocall: “—” for CGMY because path simulation is not supported.

Model	Note participation (%)	Digital (price)	BRC coupon (%)	Autocall coupon (%)
BS	63.50	5.3929	4.279	4.482
Merton	58.45	5.9031	6.118	5.561
VG	58.58	5.6905	5.835	5.673
NIG	58.27	5.7617	5.880	5.563
CGMY	56.77	5.9336	—	—
Heston	55.01	6.2922	7.772	6.572
Bates	54.80	6.3124	7.635	6.782

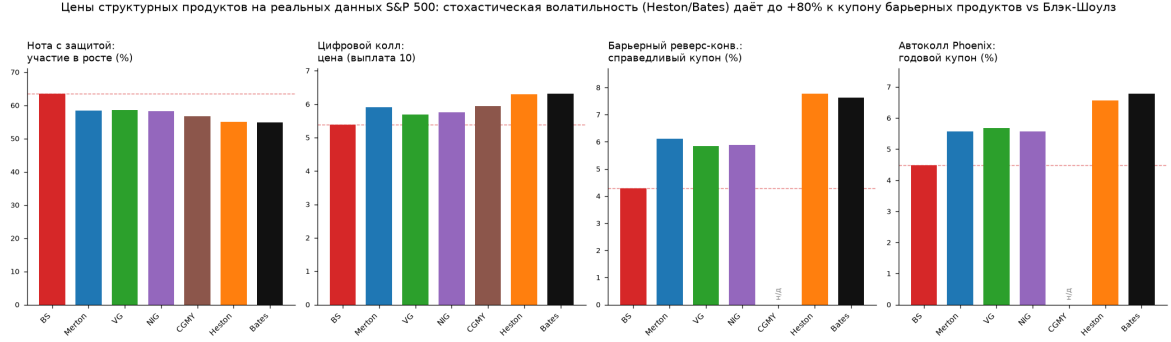


Figure 3: Prices of the four structured products across the seven models. The red dashed line is the Black–Scholes level. Stochastic-volatility models require substantially larger coupons for barrier products. Figure labels are in Russian.

6 Greeks Decomposition

6.1. Methodology

The Greeks — delta $\partial V/\partial S$, gamma $\partial^2 V/\partial S^2$, vega, and theta $-\partial V/\partial T$ — are computed by finite differences on FFT prices for each of the seven models on three instruments: ATM call, OTM put ($K/S_0 = 0.9$), and digital call. Key values for the ATM call with one-year maturity are given in Table 4.

Table 4: ATM-call Greeks (1 year) on real S&P 500 parameters.

Model	Price	Delta	Gamma	Theta
BS	617.32	0.622	3.17e−04	-377.3
Merton	667.76	0.680	2.81e−04	-428.0
VG	678.03	0.660	2.64e−04	-412.2
NIG	677.96	0.667	2.66e−04	-418.2
CGMY	675.24	0.684	2.85e−04	-437.9
Heston	693.45	0.722	2.40e−04	-462.1
Bates	692.62	0.724	2.44e−04	-463.3

6.2. Effect of Model Assumptions

The decomposition (Fig. 4) reveals a systematic effect of model structure:

- **Delta** of the ATM call rises from 0.622 (Black–Scholes) to 0.724 (Bates/Heston): a strong negative price–volatility correlation ($\rho \approx -0.735$) shifts the risk-neutral distribution through the leverage effect and increases the effective delta;
- **Gamma** differs fundamentally in shape: for all models the gamma peak is shifted left of ATM, but for Heston/Bates it is shifted furthest and is almost $1.5\times$ higher than for Black–Scholes; meanwhile the *point* gamma at ATM is, conversely, lower for them — peak and ATM gamma must not be conflated;

- **At the barrier** of the OTM put the delta divergence between models is largest: the jump model spreads breach risk across the whole smile, smoothing the local gamma profile;
- **Vega** is model-specific (different volatility parameters): comparing the raw numbers without converting to a common metric, an equivalent ATM-volatility shift, is misleading.

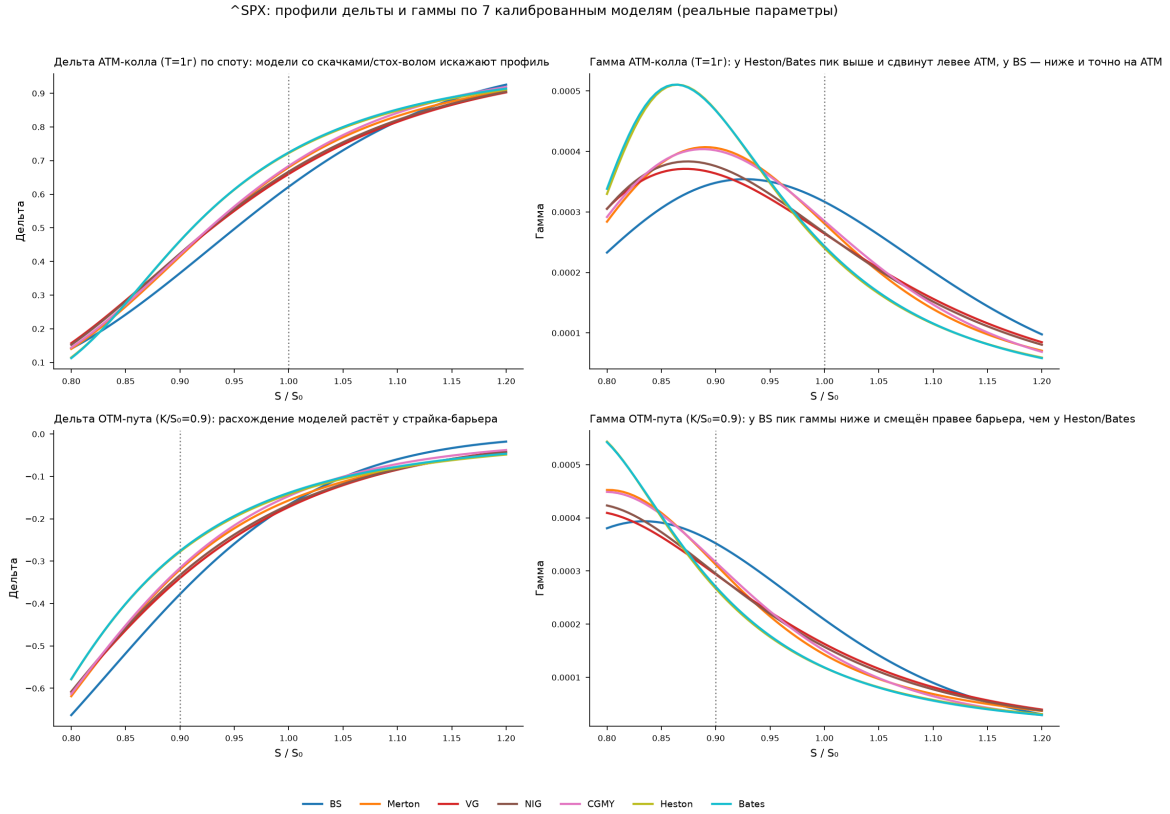


Figure 4: Delta and gamma profiles versus spot for the ATM call (top) and the OTM put at the barrier (bottom) for all seven models. Stochastic volatility and the leverage effect widen and raise the gamma peak in the drawdown region. Figure labels are in Russian.

7 Hedging-Error Analysis

7.1. Experimental Design

We simulate a discrete delta hedge of a short ATM-call position (maturity 0.25 years, weekly rebalancing, 13 steps, 30,000 paths). The “true” process (DGP) is the calibrated Bates model for the S&P 500 — a realistic process with stochastic volatility and jumps. Four hedger models (Black–Scholes, Merton, Heston, Bates) run the hedge, recomputing their delta at each step. We measure the distribution of the terminal replication P&L error (Table 5, Fig. 5).

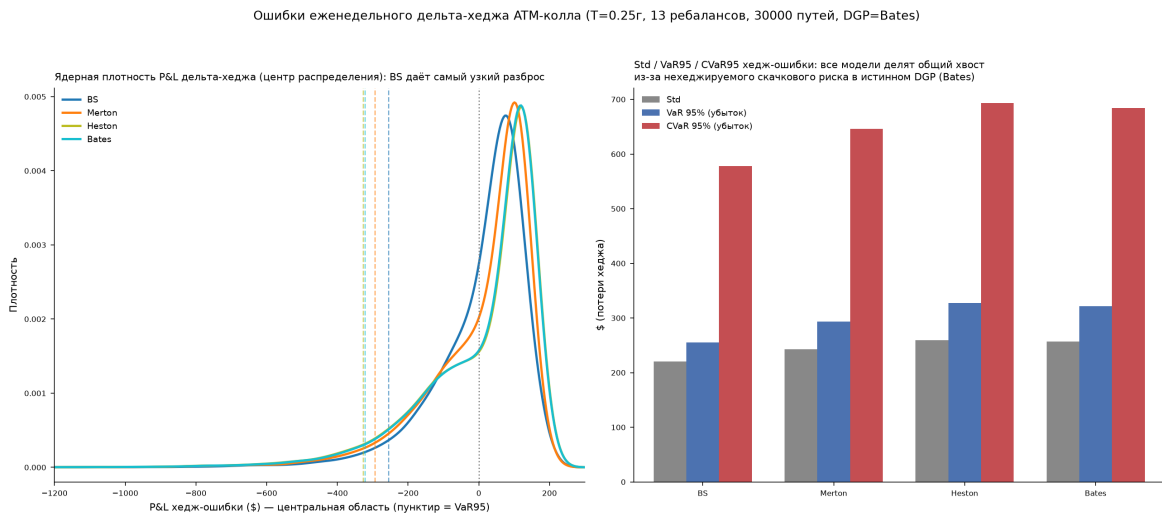
Table 5: Distribution of the delta-hedge P&L error (true process = Bates). VaR/CVaR values are loss magnitudes.

Hedger model	Std. dev.	VaR 95%	CVaR 95%	CVaR 99%	Skewness
BS	220.8	255.3	578.4	1475.0	-9.9
Merton	243.2	293.5	646.0	1619.2	-9.3
Heston	259.2	327.7	693.4	1705.0	-8.4
Bates	256.7	321.6	684.6	1689.4	-8.5

7.2. A Counterintuitive Result

The naive hypothesis that a richer model hedges better is *refuted*. The Black–Scholes hedge has the *smallest* standard deviation (220.8) and the smallest CVaR95 (578.4), whereas the Heston hedge is the widest (259.2 and 693.4). The reason is clear from Fig. 5: all hedgers share the same heavy left tail (skewness from -8.4 to -9.9 , kurtosis 113–156), generated by the unhedgeable jump risk of the true process.

Paths with a large jump (greater than 15 % in one step, approximately 2 % of paths) give an error standard deviation of approximately 1000–1120 versus approximately 130–160 without a jump. Thus **the dominant loss driver is the discreteness of rebalancing and the jump/gap risk of the process itself, not the hedger’s model choice**. Using a more complex hedging model only slightly redistributes the narrow central part of the distribution but does not remove the systemic gap risk between rebalancing dates. The observation is robust to daily rebalancing and to changing the random seed.

**Figure 5:** Left: kernel density of the hedge P&L error in the central region — Black–Scholes has the narrowest spread. Right: standard deviation, VaR95 and CVaR95 — all models share a common heavy tail due to the unhedgeable jump risk of the true process. Figure labels are in Russian.

8 Conclusions

8.1. Main Findings

1. **Fit quality.** On real smiles across all three asset classes a robust hierarchy is established: Bates (0.5–0.9 % IV-RMSE) > Heston (0.8–1.1 %) > Lévy processes (1.5–2.9 %) > Black–Scholes (4.4–6.5 %). Stochastic-volatility models are twice as accurate as pure Lévy processes thanks to the correct term structure of the smile slope.
2. **Model risk in prices.** On real data, model choice shifts the fair coupon of barrier products by *up to +82 %* (Heston/Bates versus Black–Scholes) — a discrepancy many times larger than synthetic estimates, with direct financial consequences for issuers and investors.
3. **Greeks.** Stochastic volatility, through the leverage effect, systematically raises delta and radically changes the shape of the gamma profile, shifting the sensitivity peak into the drawdown region.
4. **Hedging.** For a realistic process with jumps, the unhedgeable jump/gap risk dominates: a more complex hedger model does *not* remove the heavy left tail of the P&L. This shifts the risk-management focus from choosing the right model to managing rebalancing frequency and residual jump risk.

8.2. Limitations and Future Work

The smiles are captured at a single point in time; dynamic repeated calibration would allow assessment of parameter stability. The FX class is not covered because of the illiquidity of exchange-traded options — over-the-counter FX data would close this gap. No single-component model reproduces the steepness of very long maturities; models with jumps in volatility and rough volatility are promising. Finally, the hedging analysis can be extended to quadratic-optimal and vega hedging.

All results are obtained by real computation on live market data; reproducible code and data are saved as project artifacts.